

Computer Graphics
Project3: RollerCoaster checkpoint 3
User Manual

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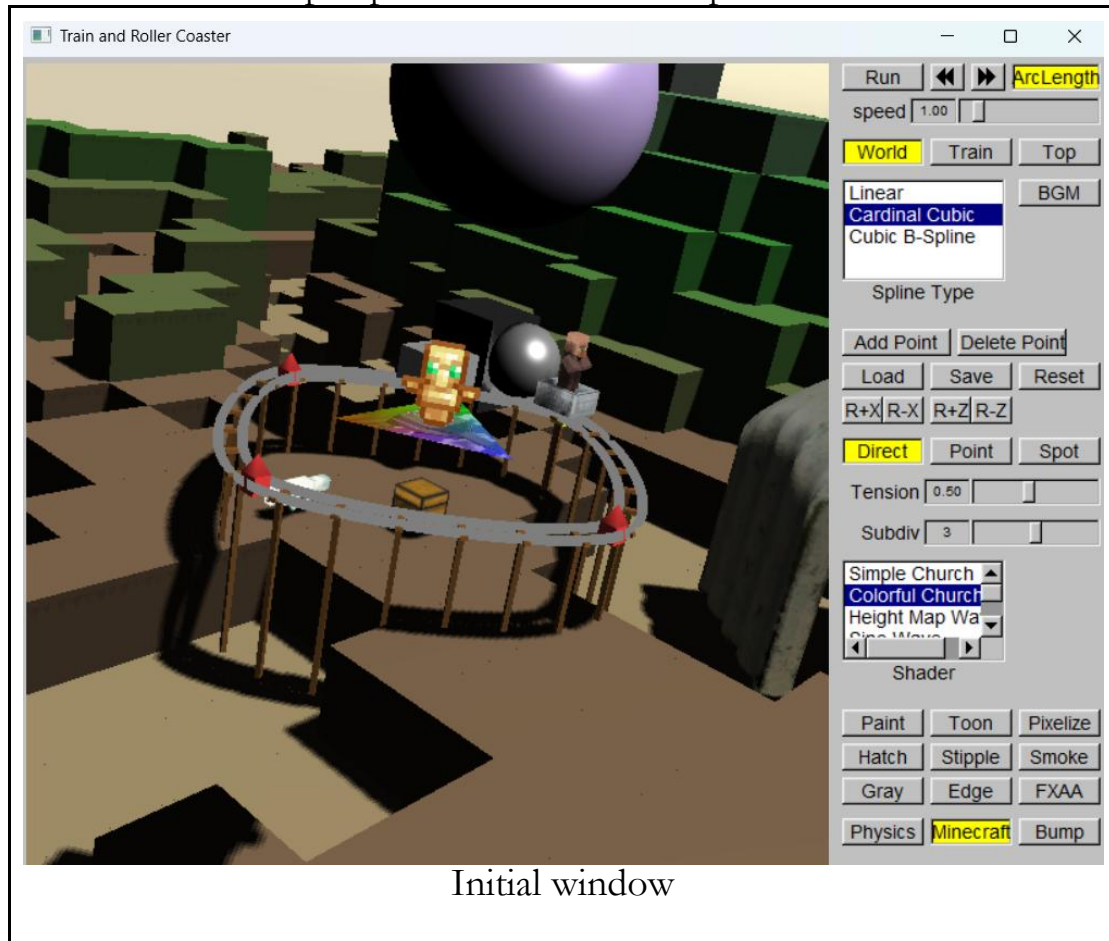
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Introduction:

RollerCoaster is a project that render a surface with a roller coaster on it.

In this project, users can use tools on the UI to change the behavior of the track and the train.

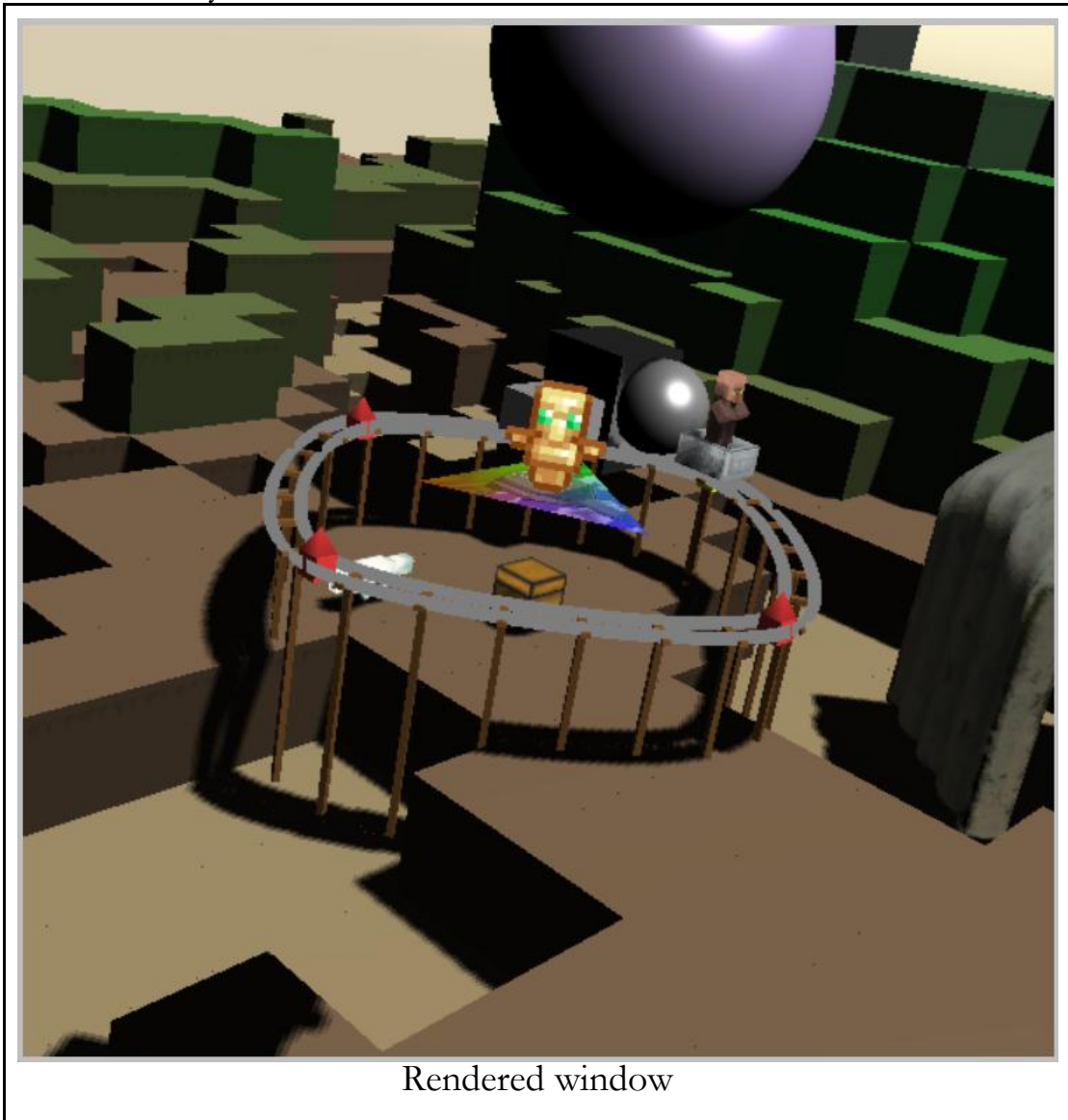
And move the perspective with mouse input.



Operations:

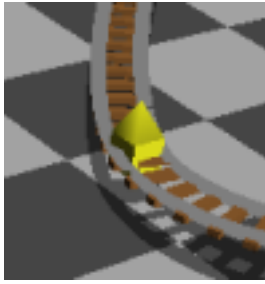
In this project, there's some operations need to be done with mouse input.

To move (rotate) the perspective, you need to right click inside the rendered window and holds it, then move the mouse with the direction you want.

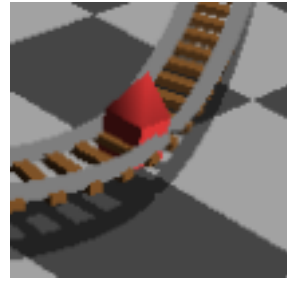


Rendered window

Next operation needs to be done with mouse input is to select the control points, you can select the control point you want with a simple left click with mouse, if a control point is selected, it's color will be changed into yellow, and the terminal will output the results at the same time.



Selected



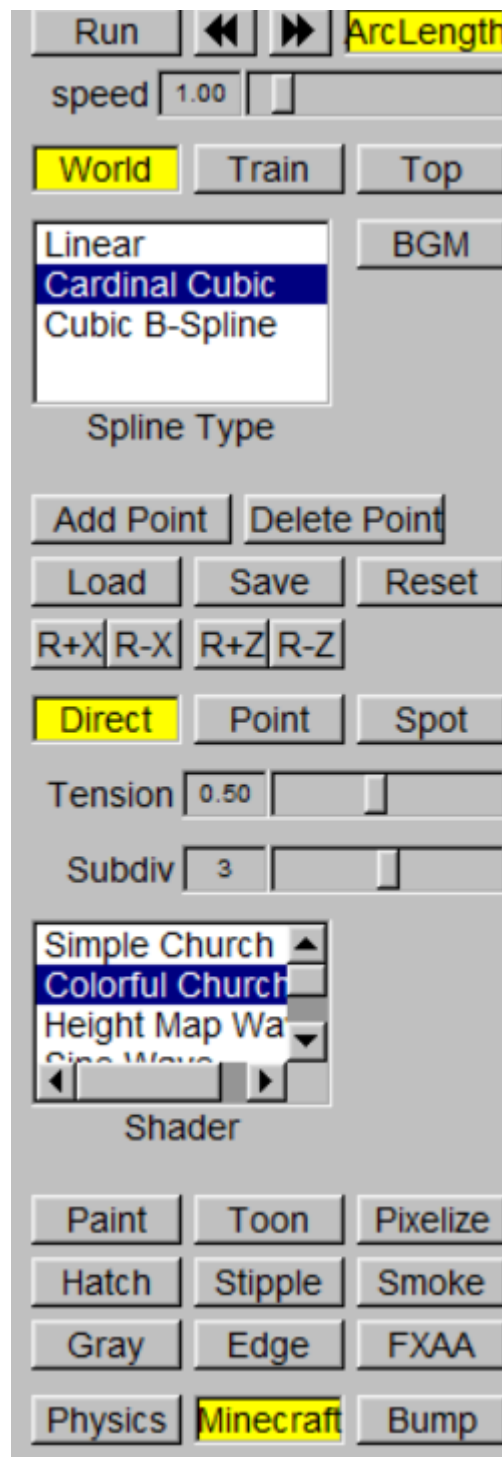
Unselected

```
C:\Users\steve\Documents\Gi X + v
CS559 Train Assignment
Selected Cube 1
Selected Cube 1
Selected Cube 2
Selected Cube 2
```

Terminal output

User Interface:

In this section, we'll show you how to use the tools on the user interface to modify the behavior of train and track.



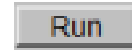
Whole user interface

Run button:

You can make train move or stop with this button on/off.



Train moves



Train stops

Fast forward/Backward button:

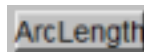
You can use these buttons to move to next/previous frame's state.



Fast forward/Backward button

ArcLength button:

With this button, you can change the move method of the train between based on time/based on arclength.



ArcLength button

Speed slider:

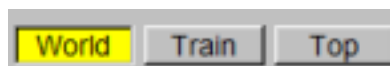
With this tool, you can adjust the speed of the moving train.



Speed slider

Camera button group:

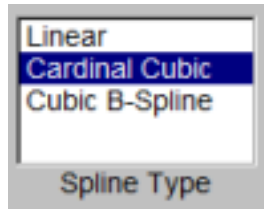
You can change the camera with world/train/top view with this button group.



Camera button group

Spline browser:

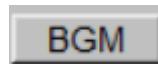
Similar as previous tool, you can change the spline mode in between these three types Linear/Cardinal Cubic/Cubic B-spline.



Spline browser

BGM button:

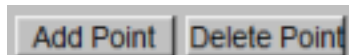
With this button, you can choose to play or not to play the BGM, there's two different music in this project.



BGM button

Add/Delete control points button:

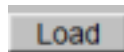
Same as their name, you can add/delete control points with these buttons.



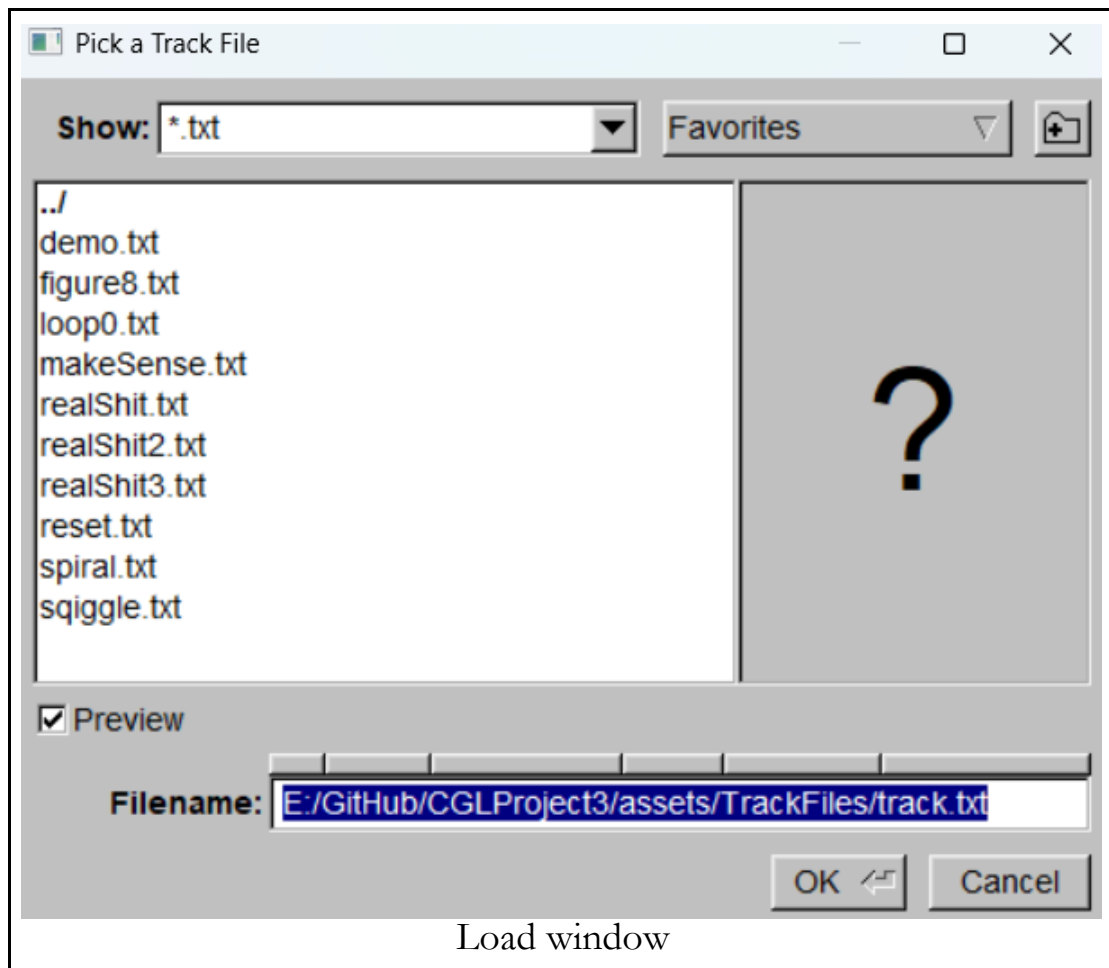
Add/Delete control points button

Load button:

You can load the generated track file with this button.

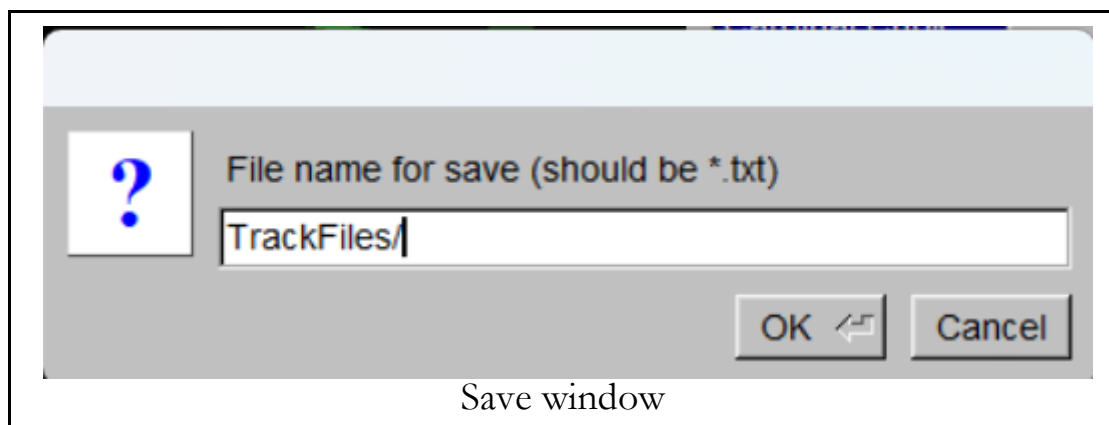
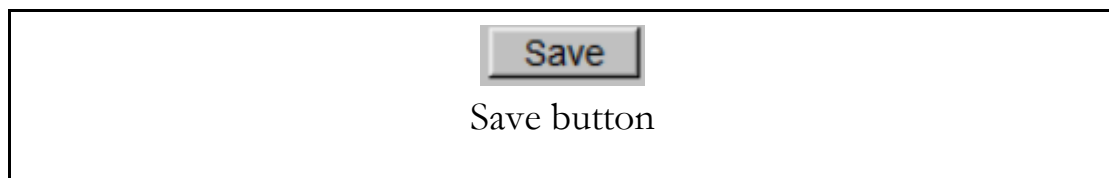


Load button



Save button:

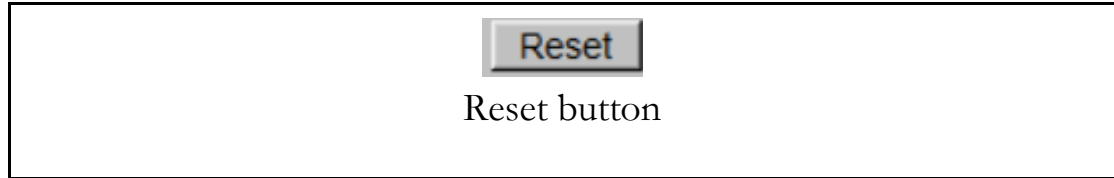
You can save the track that is current rendered on your screen with this button.



Remember that your track file must be a .txt file.

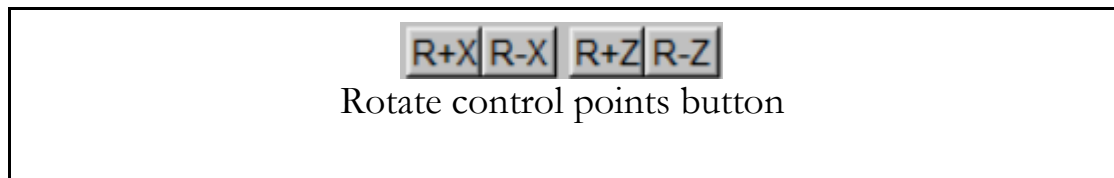
Reset button:

You can reset the track back to initial condition with this button.



Rotate control points button:

You can rotate the selected control point's x/z coordinate with these buttons.



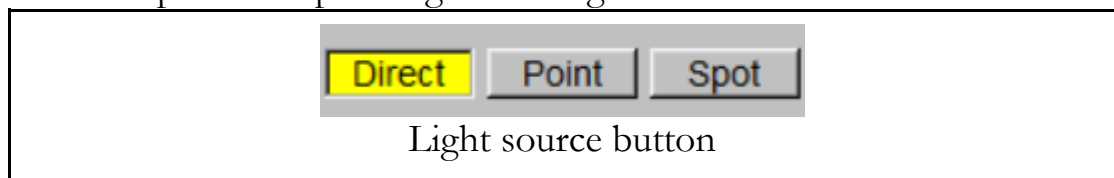
Light source button:

With these buttons, you can enable the light sources you want, and it's additive.

Direct corresponding to directional light source.

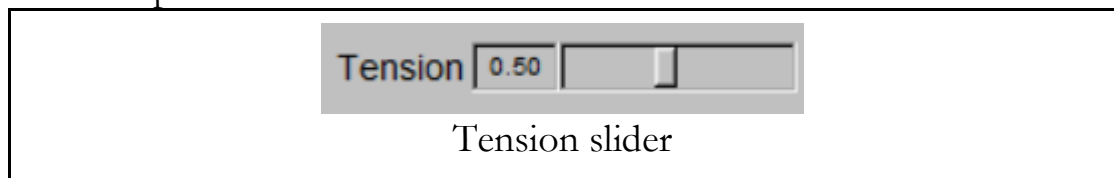
Point corresponding to point light source.

Spot corresponding to headlights of train.



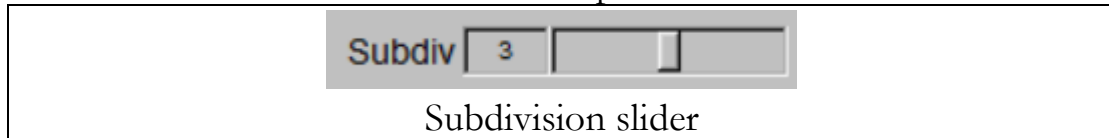
Tension slider:

With this slider, you can control the tension of Cardinal Cubic spline.



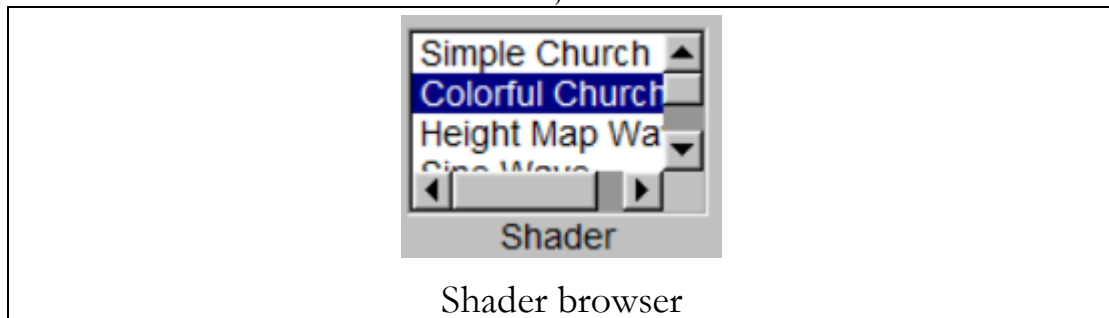
Subdivision slider:

With this slider, you can control the times you want to do the subdivision with the sphere.



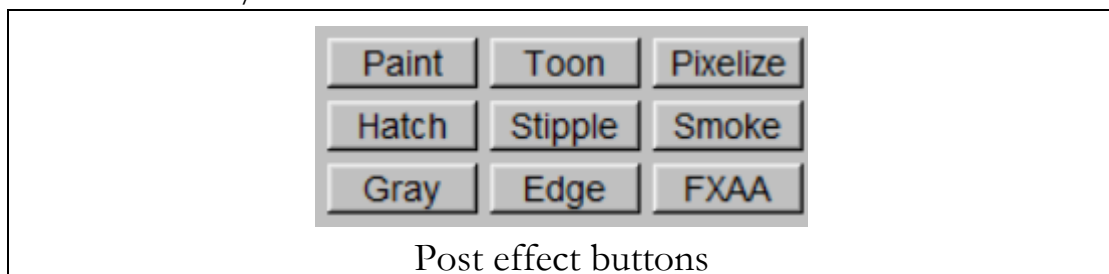
Shader browser:

With this browser, you can choose which shader you want to use to render the objects in the middle of the screen.



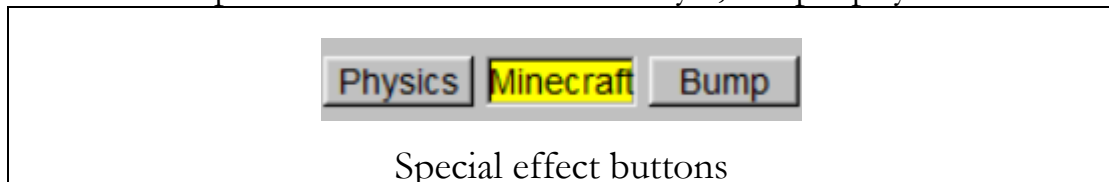
Post effect buttons:

With these buttons, you can choose whatever post effect you want (e.g. NPR Paint, Toon shading...) and to turn on/off smoke effect.



Special effect buttons:

With these buttons, you can enable/disable some of our special effects like Minecraft style, Simple physics...



Little game:

After you hit the run button, you can see the ghaſt from Minecraft / fighter jet (with Minecraft button off) is moving for random directions, and the ghaſt / fighter jet will fire a 'TNT' forward to the train, you need to protect the train with the left click of your mouse, otherwise the train will be reset to the initial control point.



The ghaſt



Fighter jet



The TNT